Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

CN^- is a strong field ligand. This is due to the fact that

(A) It can accept electron from metal species

(B) It forms high spin complexes with metal species

(C) It carries negative charge

(D) It is a pseudohalide

Q.2

Which one of the following complexes is an outer orbital complex
Atomic nos : $Mn = 25, Fe = 26, Co = 27, Ni = 28$

(A) $[Co\,(NH_3)_3]^{3+}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{4-}$

(D) $[Ni\,(NH_3)_6]^{2+}$

Q.3

In the complex $[SbF5]^{2-}$, $sp^3\ d$ hybridisation is present Geometry of the complex is

(A) Square pyramidal

(B) Square bipyramidal

(C) Tetrahedral

(D) Square

Q.4

An anion solution gives a white ppt with $AgNO_3$ solution. The ppt. dissolves in dil. ammonia due to the formation of

(A) $AgNO_3$

(B) NH_4NO_3

(C) $\lfloor Ag\,(NH_3)_2 \rfloor Cl$

(D) $\lfloor Ag\,(NH_3)_2 \rfloor Br$

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Q.5

Which of the following is incorrect regarding spectrochemical series?

(A) $NH_3 > H_2O$

(B) $F^- > C_2O_4^{2-}$

(C) $NCS^- > SCN^-$

(D) en $>$ EDTA

Q.6

Nickel $(Z = 28)$ combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively:

(A) one, square planar

(B) two, square planar

(C) one, tetrahedral

(D) two, tetrahedral

Q.7

Which of the following are inner orbital complex (i.e., involving d^2sp^3 hybridisation) and is paramagnetic in nature?

(A) $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$

(B) $[MnCl_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$

(C) $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$

(D) $[MnCl_6]^{3-}$, $[Fe(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$

Q.8

Which of the following is organo-metallic compound?

(A) $Ti(C_2H_4)_4$

(B) $Ti(OC_2H_5)_4$

(C) $Ti(OCOCH_3)_4$

(D) $Ti(OC_6H_5)_4$

Q.9

The structure of which of the following chloro species can be explained on the basis of dsp^2 hybridization ?

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(A) $PdCl_4^{2-}$

(B) $FeCl_4^{2-}$

(C) $CoCl_4^{2-}$

(D) $NiCl_4^{2-}$

Q.10

$\left[NiCl_2\left\{P\left(C_2H_5\right)_2\left(C_6H_5\right)\right\}_2\right]$ exhibits temperature dependent magnetic behaviour (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively

(A) tetrahedral and tetrahedral

(B) square planar and square planar

(C) tetrahedral and square planar

(D) square planar and tetrahedral

Q.11

Which of the following complex compound(s) is/are paramagnetic and low spin?
(I) $K_3\left[Fe\left(CN_6\right)\right]$
(II) $\left[Ni(CO)_4\right]^0$
(III) $\left[Cr\left(NH_3\right)_6\right]^{3+}$
(IV) $\left[Mn(CN)_6\right]^{4-}$
Choose the correct code:

(A) I only

(B) II and III

(C) I and IV

(D) IV only

Q.12

An aqueous solution of titanium chloride, when subjected to magnetic measurement, measured zero magnetic moment. Assuming the octahedral complex in aqueous solution, the formulae of the complex is:

(A) $\left[Ti\left(H_2O\right)_6\right]Cl_2$

(B) $\left[Ti\left(H_2O\right)_6\right]Cl_4$

(C) $\left[TiCl_3\left(H_2O\right)_3\right]$

(D) $\left[TiCl_2\left(H_2O\right)_4\right]$

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Q.13

Which of the following statement is not true for the reaction given below?
$\left[\mathrm{Cu}\left(\mathrm{H}_2\mathrm{O}\right)_4\right]^{2+} + 4\mathrm{NH}_3 \rightleftharpoons \left[\mathrm{Cu}\left(\mathrm{NH}_3\right)_4\right]^{2+} + 4\mathrm{H}_2\mathrm{O}$

(A) It is a ligand substitution reaction

(B) NH_3 is relatively strong field ligand while $\mathrm{H}_2\mathrm{O}$ is a weak field ligand

(C) During the reaction, there is a change in colour form light blue to dark blue

(D) $\left[\mathrm{Cu}\left(\mathrm{NH}_3\right)_4\right]^{2+}$ has tetrahedral structure and is paramagnetic

Q.14

The d -electron configurations of $\mathrm{Cr}^{2+}, \mathrm{Mn}^{2+}, \mathrm{Fe}^{2+}$ and Co^{2+} are d^4, d^5, d^6 and d^7 respectively. Which one of the following will exhibit the lowest paramagnetic behaviour? (Atomic no. $\mathrm{Cr}=24, \mathrm{Mn}=25, \mathrm{Fe}=26, \mathrm{Co}=27$).

(A) $\left[\mathrm{Co}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$

(B) $\left[\mathrm{Cr}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$

(C) $\left[\mathrm{Mn}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$

(D) $\left[\mathrm{Fe}\left(\mathrm{H}_2\mathrm{O}\right)_6\right]^{2+}$

Q.15

The shape of $\left[\mathrm{Cu}\left(\mathrm{NH}_3\right)_4\right]^{2+}$ is

(A) tetrahedral

(B) square planar

(C) pyramidal

(D) octahedral

Q.16

Which of the following statements is incorrect?

(A) $\left[\mathrm{MnCl}_6\right]^{3-}$ is more paramagnetic than $\left[\mathrm{Mn}(\mathrm{CN})_6\right]^{3-}$

(B) Both $\left[\mathrm{Co}\left(\mathrm{C}_2\mathrm{O}_4\right)_3\right]^{3-}$ and $\left[\mathrm{CoF}_6\right]^{3-}$ are paramagnetic.

(C) $\left[\mathrm{Fe}(\mathrm{CN})_6\right]^{3-}$ forms inner orbital complex whereas $\left[\mathrm{FeF}_6\right]^{3-}$ forms outer orbital complex.

(D) Both (a) and (b).

Q.17

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An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is:

(A) sp^3d^2

(B) dsp^2

(C) d^2sp^3

(D) sp^3d

Q.18

The correct statement about the magnetic properties of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ is $(Z = 26)$:

(A) both are paramagnetic.

(B) both are diamagnetic.

(C) $[Fe(CN)_6]^{3-}$ is diamagnetic, $[FeF_6]^{3-}$ is paramagnetic.

(D) $[Fe(CN)_6]^{3-}$ is paramagnetic, $[FeF_6]^{3-}$ is diamagnetic.

Q.19

Nickel $(Z = 28)$ combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiX_4]^{2-}$. The hybridization involved and the number of unpaired electrons present in the complex is respectively:

(A) sp^2, two

(B) dsp^2, zero

(C) dsp^2, one

(D) sp^3, zero

Q.20

Prussian blue is a deep blue pigment containing Fe^{2+}, Fe^{3+} and CN^- ions. It has the formula Fe_7CN_{18}. How many Fe^{2+} and Fe^{3+} ions are there per formula unit?

(A) $3\,Fe^{2+}$ and $4\,Fe^{3+}$

(B) $4\,Fe^{2+}$ and $3\,Fe^{3+}$

(C) $5\,Fe^{2+}$ and $2\,Fe^{3+}$

(D) $6\,Fe^{2+}$ and $1\,Fe^{3+}$

Answers & Solutions

Q.1 Answer:

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It is a pseudohalide

Solution:

Cyanide ion is strong field ligand because it is a pseudohalide ion, pseudohalide ions are stronger coordinating ligand & they have the ability to form σ bond from pseudohalide to the metal and π bond (from the metal to pseudohalide).

Q.2 Answer:

$[\mathrm{Ni}\,(\mathrm{NH_3})_6]^{2+}$

Solution:

Complex Hybridization of central atom $[\mathrm{Re(CN)_6}]^4$ $\mathrm{d^2\ sp^3}$ **(inner)**

$[\mathrm{Mn}(CN)_6]^4$ d^2sp^3 **(inner)**

$[\mathrm{Co}\,(\mathrm{NH_3})_6]^{3+}$ $d^2\mathrm{sp^3}$ **(inner)**

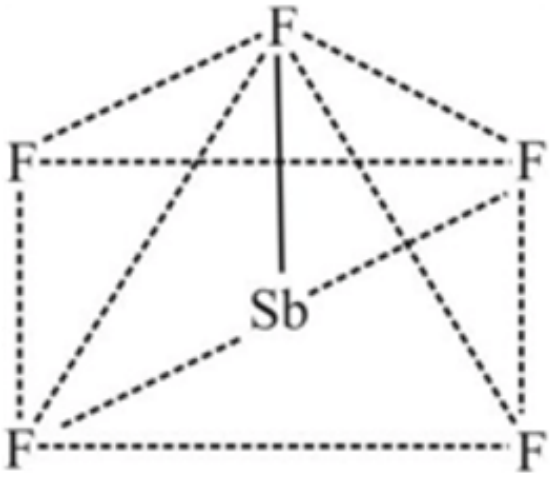
$[\mathrm{Ni}\,(\mathrm{NH_3})_6]^{2+}$ $\mathrm{sp^3\ d^2}$ **(outer)**

Q.3 Answer:

Square pyramidal

Solution:

$[\mathrm{SbF_5}]^{2-}$ **has** sp^3d **hybridization and shows square pyramidal geometry**



Q.4 Answer:

$[\mathrm{Ag}\,(\mathrm{NH_3})_2]\,\mathrm{Cl}$

Solution:

$$\mathrm{AgNO_3 + Cl^- \rightarrow AgCl + NO_3^-}$$
$$\mathrm{AgCl + 2NH_3 \rightarrow [Ag\,(NH_3)_2]\,Cl}$$

Q.5 Answer:

$\mathrm{F^- > C_2O_4^{2-}}$

Solution:

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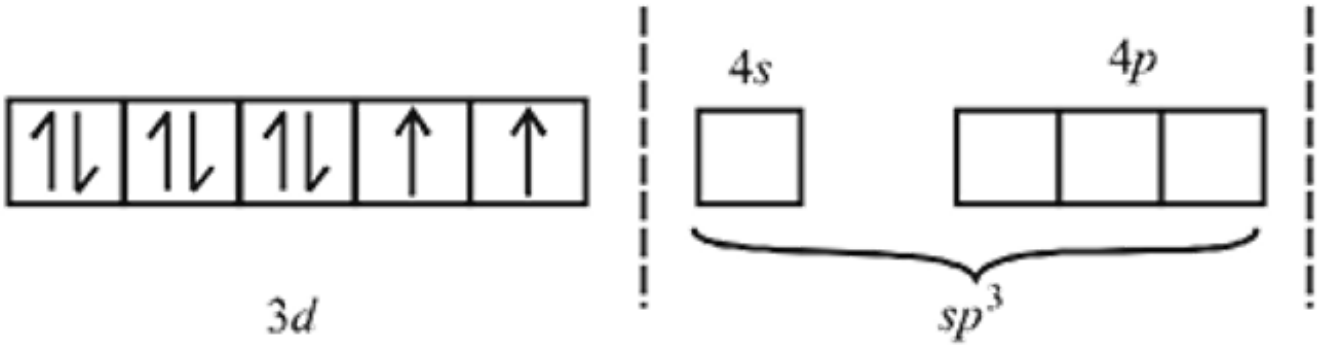
According to spectrochemical series $C_2O_4^{2-} > F^-$

Q.6 Answer:

two, tetrahedral

Solution:

$[NiX_4]^{2-}$, the electronic configuration of Ni^{2+} is



It contains two unpaired electrons and the hybridisation is sp^3 (tetrahedral).

Q.7 Answer:

$[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$

Solution:

$[Mn(CN)_6]^{3-}$ and $[Fe(CN)_6]^{3-}$ are inner orbital complexes and paramagnetic while $[Co\,(C_2O_4)_3]^{3-}$ is diamagnetic in nature.

Q.8 Answer:

$Ti\,(C_2H_4)_4$

Solution:

$Ti\,(C_2H_4)_4$ is an organometallic compound due to Ti directly attached to C- atom

Q.9 Answer:

$PdCl_4^{2-}$

Solution:

$[PdCl_4]^{2-}$ is dsp^2 hybridized and square planar in shape.

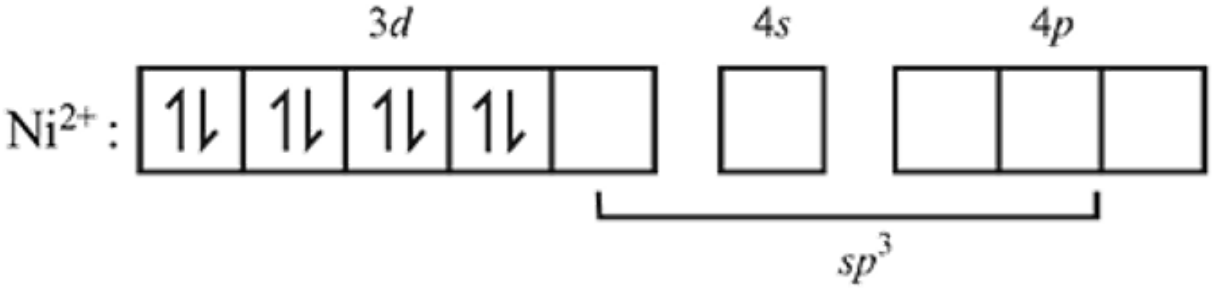
Q.10 Answer:

tetrahedral and square planar

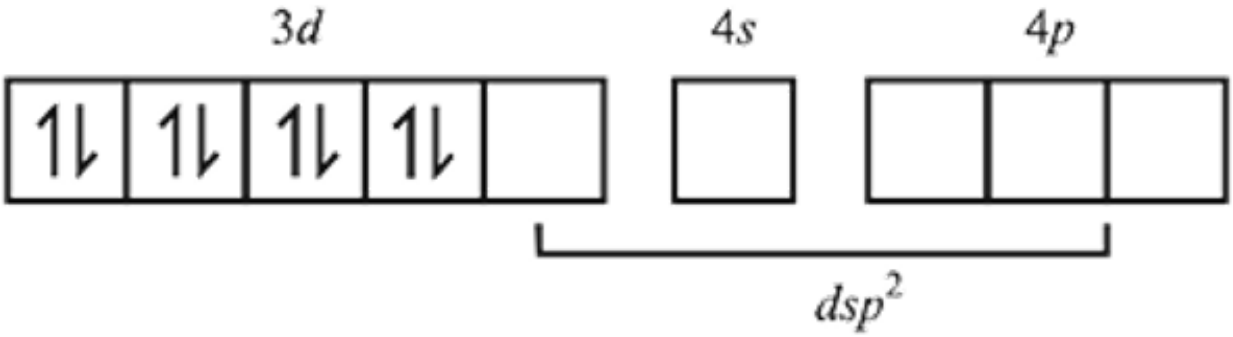
Solution:

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In both states (paramagnetic and diamagnetic) of the given complex, Ni exists as Ni^{2+} whose electronic configuration is $[Ar]3d^8 4s^0$.



In the above paramagnetic state the geometry of the complex is sp^3 giving tetrahedral geometry. The diamagnetic state is achieved by pairing of electrons in $3d$ orbital.



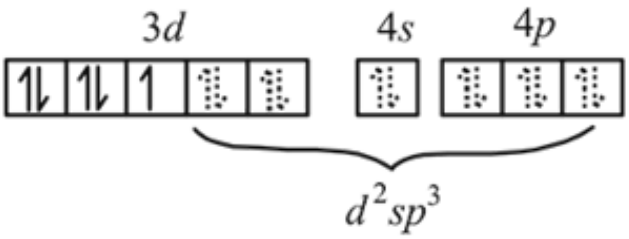
Thus the geometry of the complex will be dsp^2 giving square planar geometry.

Q.11 Answer:

I and IV

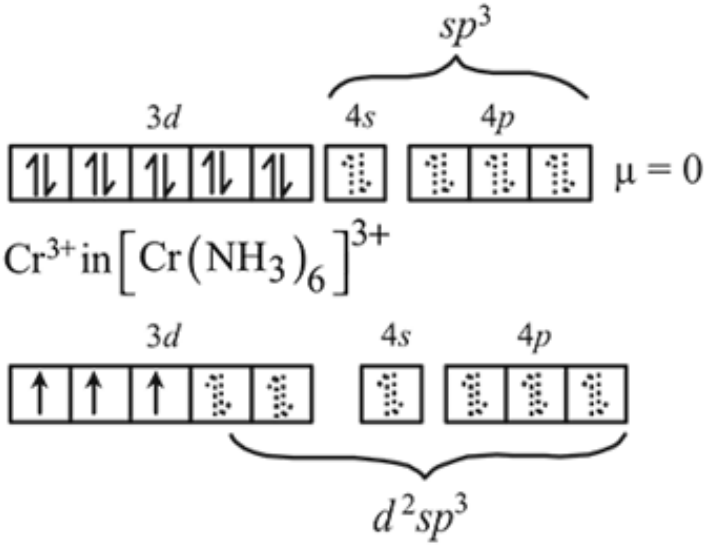
Solution:

Fe^{3+} **in** $[Fe(CN_6)]^{3-}$



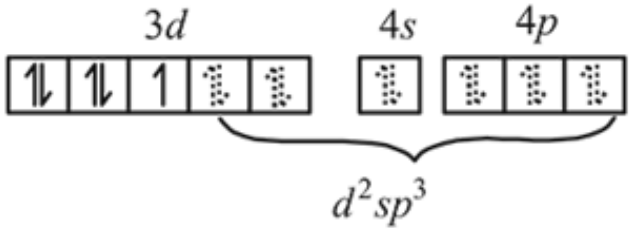
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$\Rightarrow$ **low spin complex** $\mu = 1.732\text{BM}$ **Ni in** $[\text{Ni(CO)}_4]$



$\Rightarrow$ **low and high spin complex is applicable for** d^4 **to** d^7 **configuration**

Mn^{2+} **in** $[\text{Mn(CN)}_6]^{4-}$



$\Rightarrow$ **low spin complex** $\mu = 1.732\text{BM}$

Q.12 Answer:

$[\text{Ti}\,(\text{H}_2\text{O})_6]\,\text{Cl}_4$

Solution:

$[\text{Ti}\,(\text{H}_2\text{O})_6]\,\text{Cl}_4$

Coordination number 6 $\Rightarrow$ **octahedral complex Ti is in** $+4$ **oxidations state** $\Rightarrow$ **no unpaired electrons**

$\Rightarrow$ **magnetic moment** $= 0$ **B.M.**

Q.13 Answer:

$[\text{Cu}\,(\text{NH}_3)_4]^{2+}$ has tetrahedral structure and is paramagnetic

Solution:

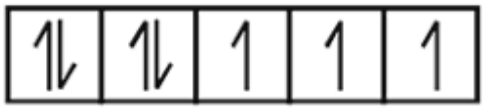
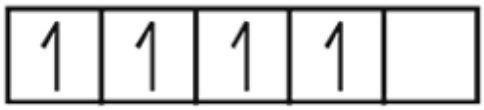
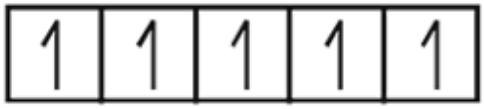
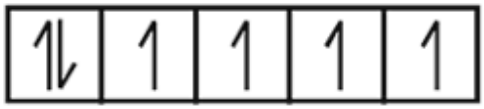
$[\text{Cu}\,(\text{NH}_3)_4]^{2+}$ has square planar structure and is paramagnetic.

Q.14 Answer:

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$[Co(H_2O)_6]^{2+}$

Solution:

	Electronic configuration		No. of unpaired electrons
Co^{2+}	d^7		3
Cr^{2+}	d^4		4
Mn^{2+}	d^5		5
Fe^{2+}	d^6		4

$\therefore$ Since Co^{3+} has lowest no. of unpaired electrons hence lowest paramagnetic behaviour is shown by $[Co(H_2O)_6]^{2+}$

Q.15 Answer:

square planar

Solution:

Shape of $[Cu(NH_3)_4]^{2+}$ is square planar.

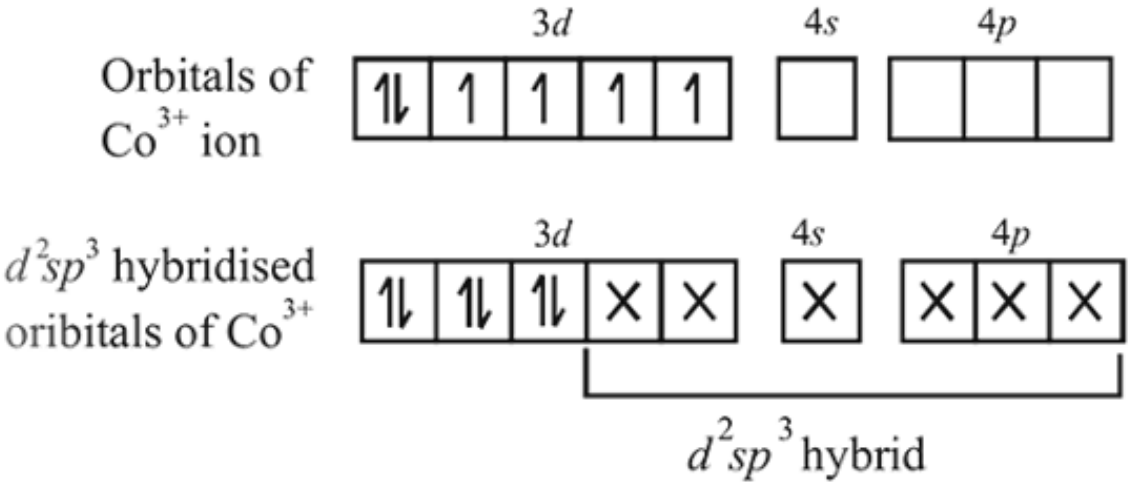
Q.16 Answer:

Both $[Co(C_2O_4)_3]^{3-}$ and $[CoF_6]^{3-}$ are paramagnetic.

Solution:

$[Co(C_2O_4)_3]^{3-}$ is diamagnetic as oxalate is a strong ligand causing pairing of $3d$ electrons in Co^{3+} thereby leading to d^2sp^3 hybridisation.

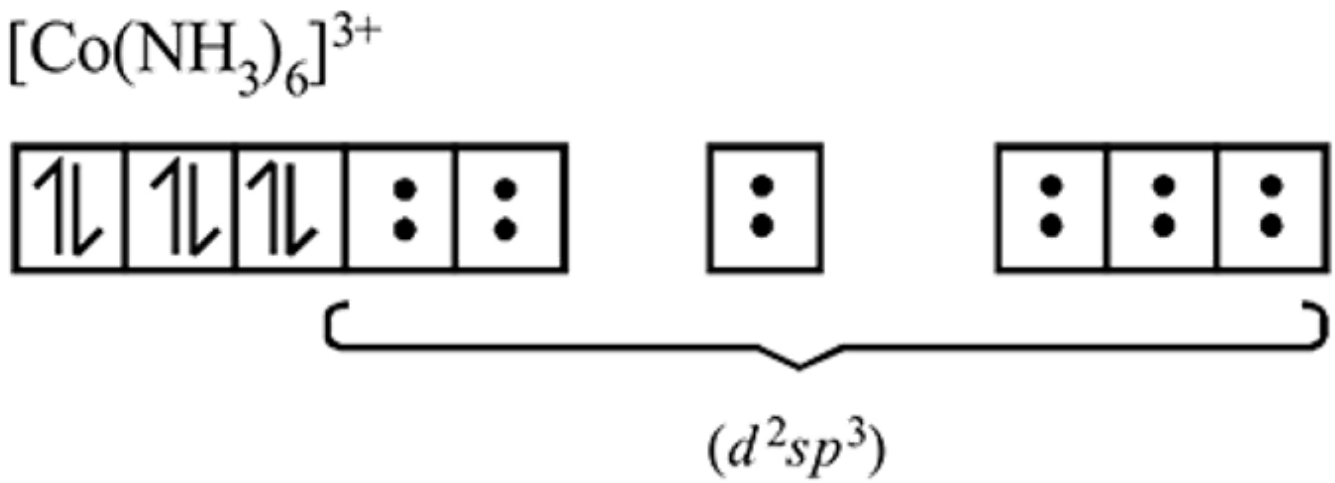
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Q.17 Answer:

d^2sp^3

Solution:



Octahedral and Diamagnetic

Q.18 Answer:

both are paramagnetic.

Solution:

Both are paramagnetic, the only difference is that CN^- is a strong field ligand whereas F^- is a weak field ligand.

Q.19 Answer:

dsp^2, zero

Solution:

Monodentate ligand is uninegative and hence, the oxidation number of Ni in $[NiX_4]^{2-}$ is $+2$.

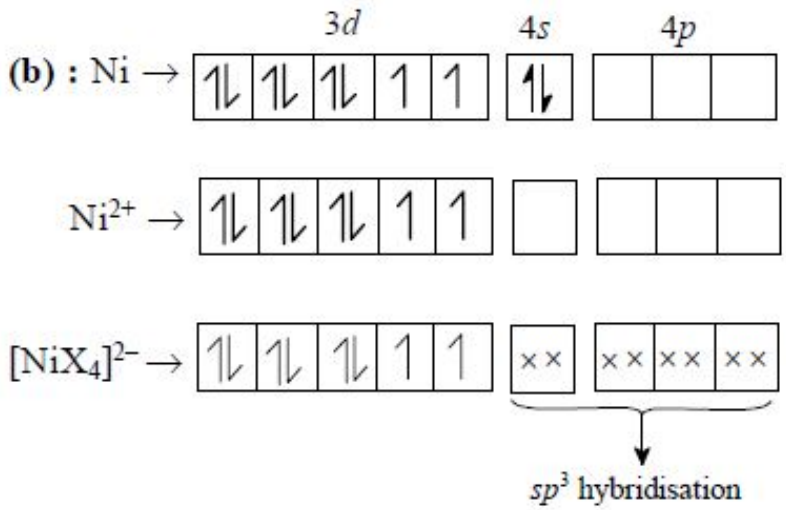
The electronic configuration of Ni^{2+} is $[Ar]3\ d^8$.

If the compound is diamagnetic it means all the electrons are paired. Hence, there will be one 3 d

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orbital empty and one 4 s orbital empty.

The hybridisation involved in the formation of complex $[NiL_4]^{2-}$ is dsp^2.



Number of unpaired electrons = 2

Geometry = tetrahedral.

Q.20 Answer:

$3Fe^{2+}$ and $4Fe^{3+}$

Solution:

Prussian blue is a deep blue pigment is Fe_4FeCN_{63}

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